Geodesic Graph Neural Network for Efficient Link Prediction

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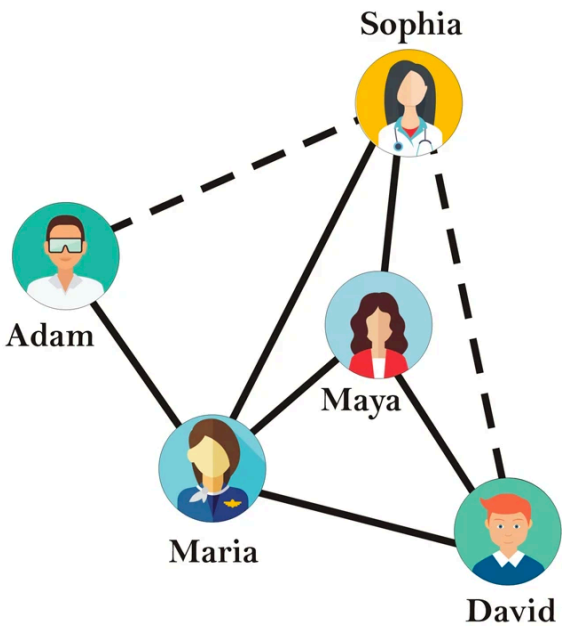
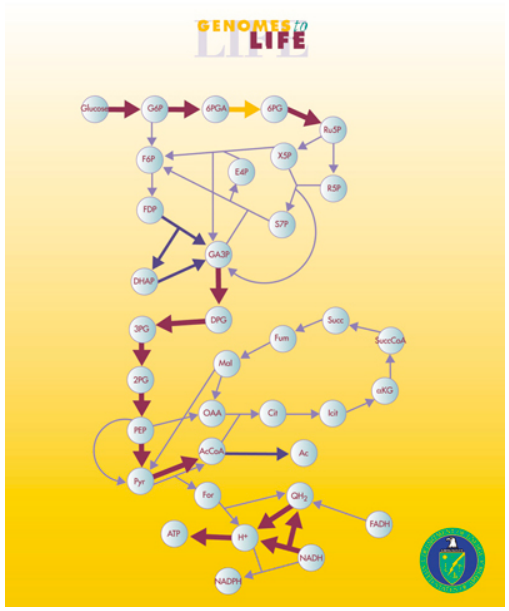
CSE Department DSS Talk

Link prediction is essential in graph learning

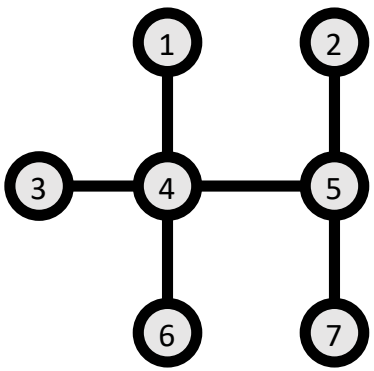


Social Network Recommender System



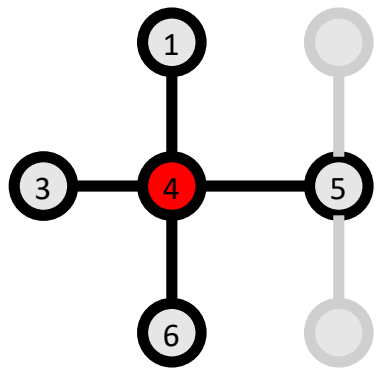
Metabolic Network Reconstruction

Graph Neural Network is a popular solutions



Each node has an initial features

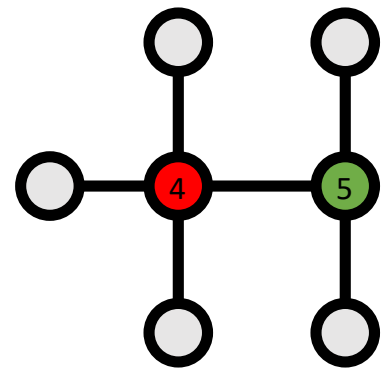
e_i^0



Red node read 1,3,5,6 from the neighbors, and aggregate them.

$a_4 = AGG(E_{[1,3,5,6]})$

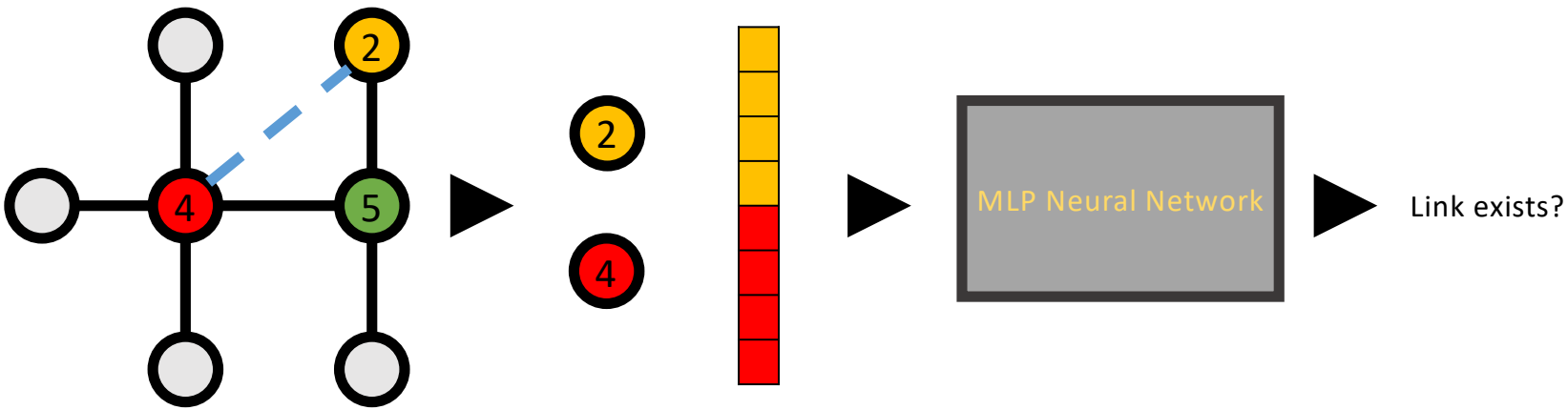
$e_4^1 = COMBINE(a_4, e_4^0)$



Such operation is applied to every node in one GNN layer, and many layers are ***stacked*** together

$E = \{e_1^k, \dots, e_7^k\}$

Graph Neural Network is a popular solutions



Predict if the blue dotted link exists

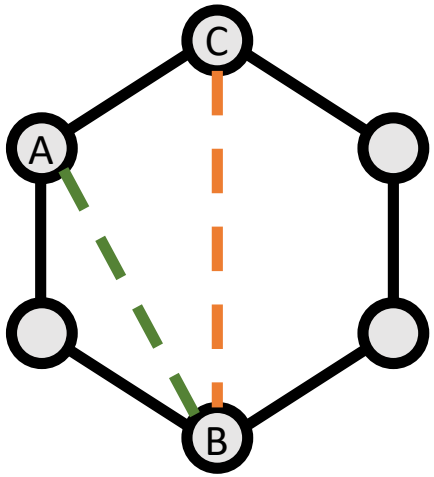
Node Vectors/Embeddings are extracted

$h_{2,4} = e_2 \oplus e_4$

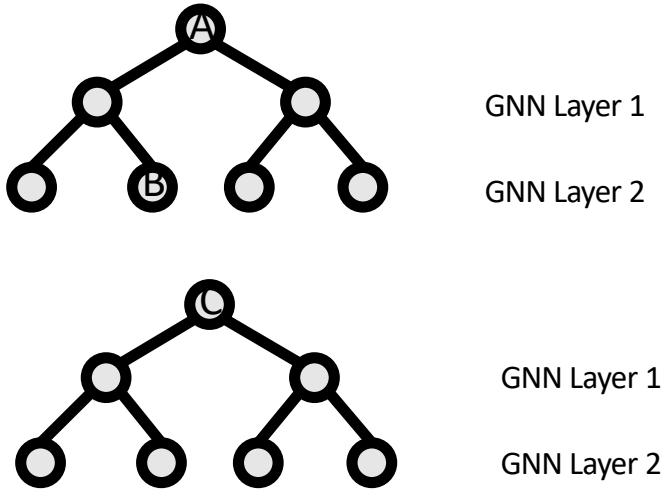
A MLP Neural Network Maps the vector to the probability of the link

$p = MLP(h_{2,4})$

Direct GNN is not good enough

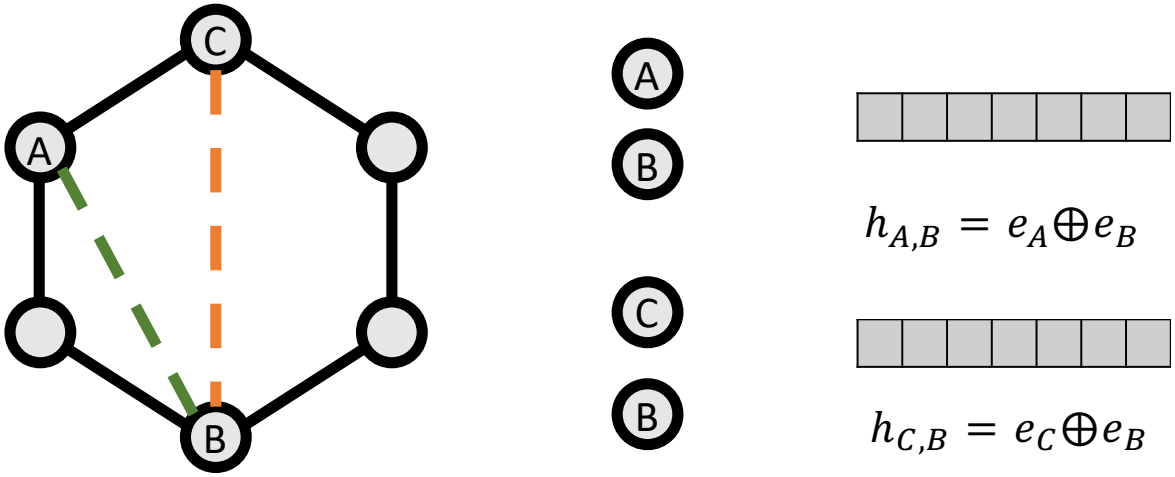


Node A and Node C are symmetric in the graph, assuming all nodes start with the same feature



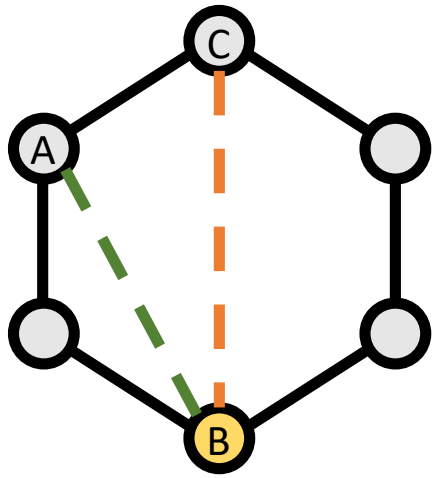
The computation graph of A and C are the same

Direct GNN is not good enough

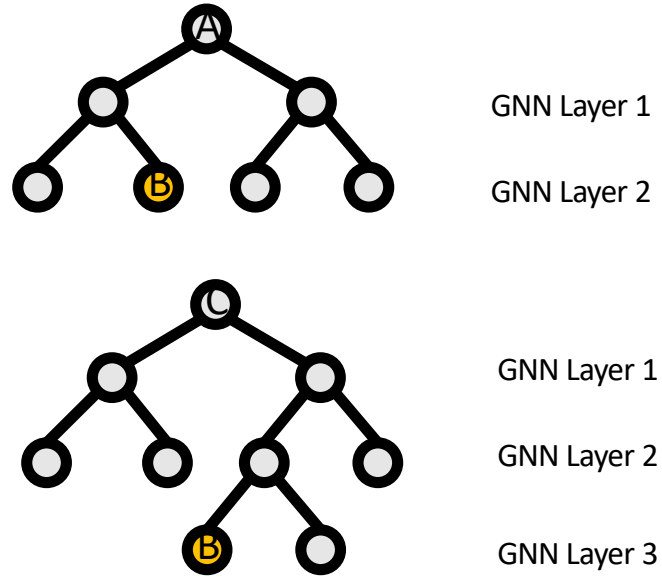


$$e_A = e_C \rightarrow h_{A,B} = h_{C,B}$$

Direct GNN is not good enough

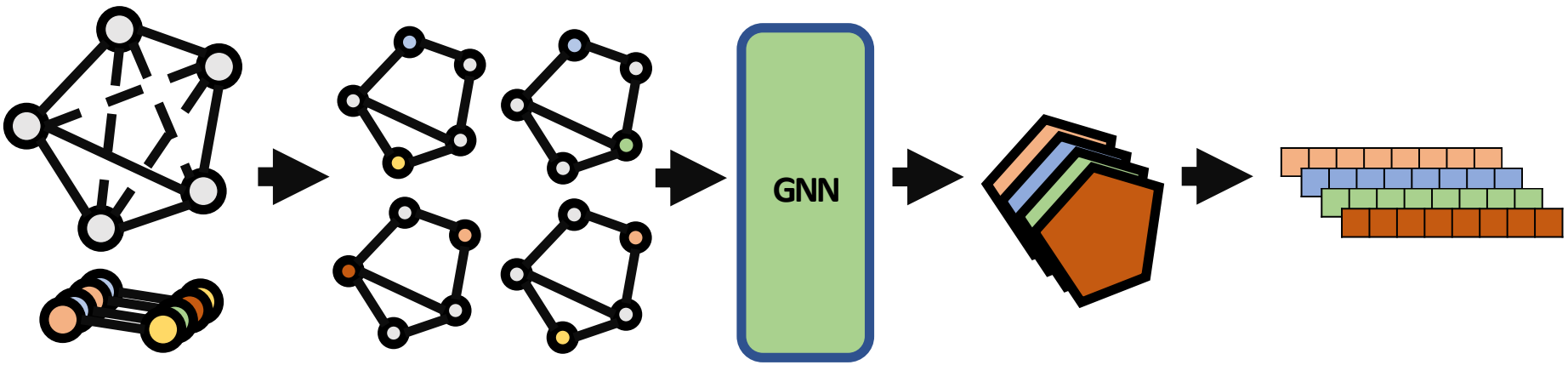


Node A and Node C are symmetric in the graph, but the node of interest B is labelled with a different feature



The labelled nodes distinguish the computation graphs of A and C

Labeling methods are very slow

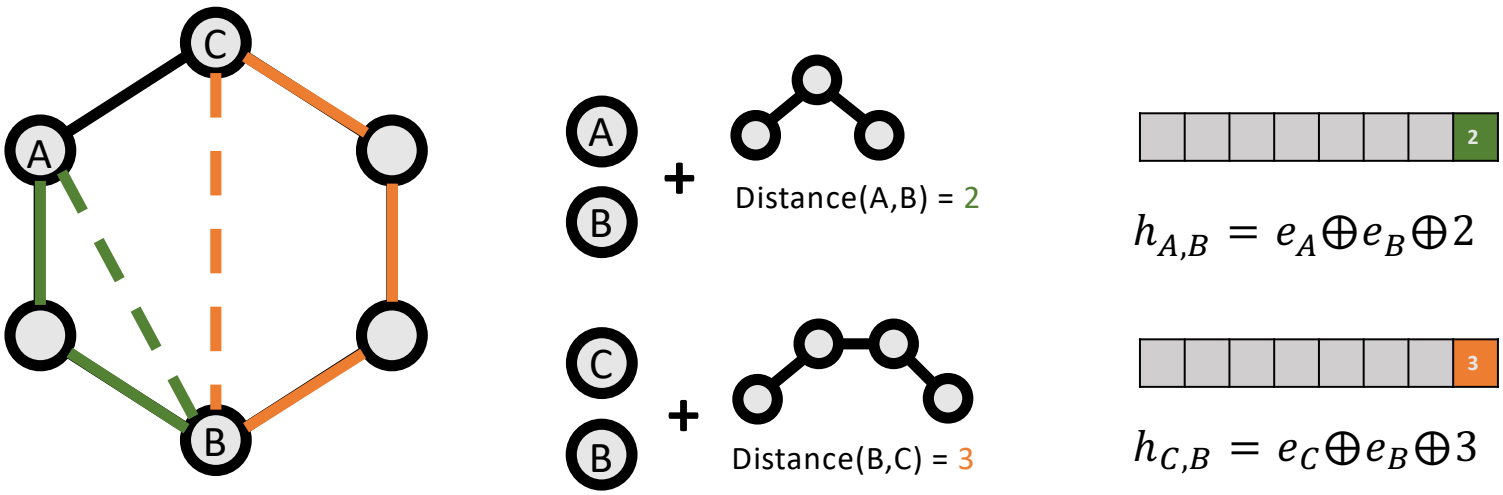


Node labeling results in multiple graphs

GNN applied to all graphs

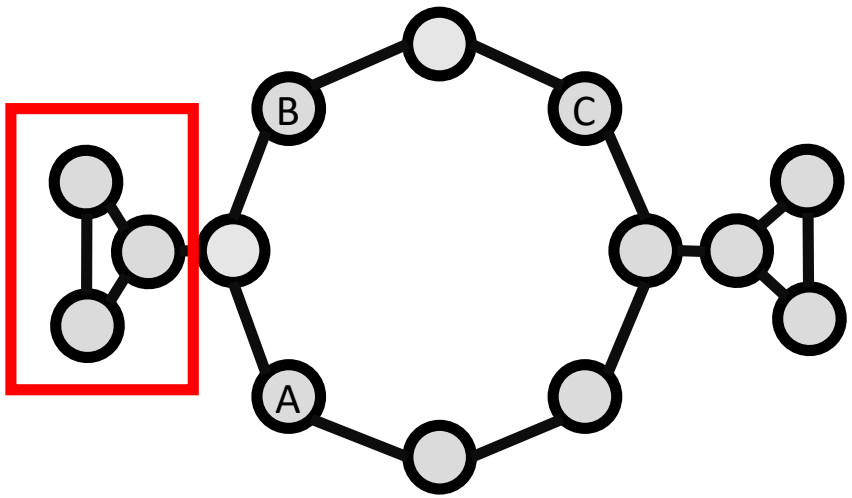
Node Vectors are then extracted

Geodesic Information



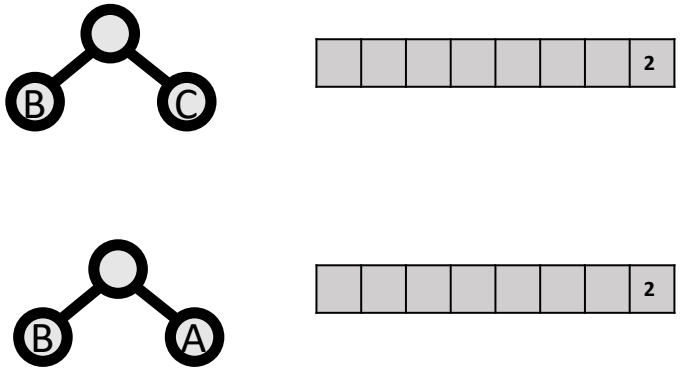
$h_{A,B} \neq h_{C,B}$, AB and CB are distinguished

Neural Geodesic > Shortest Path



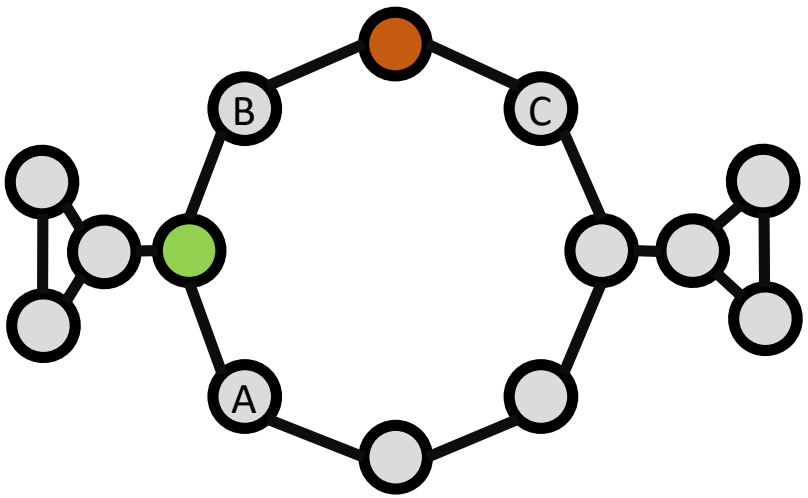
A and C are symmetric

Dist(B,C) and Dist(A,B) are both 2

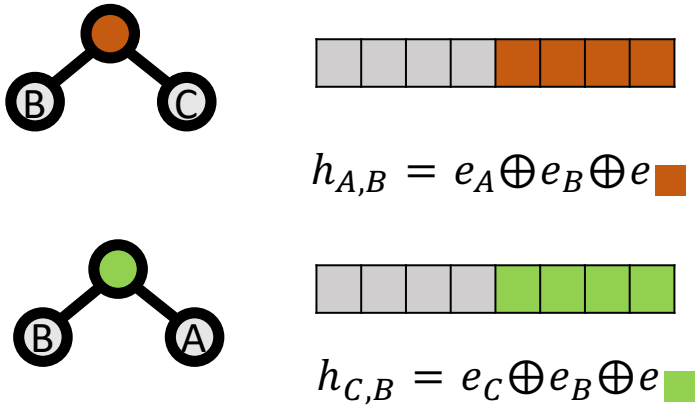


Using the only the shortest path is not able to distinguish the two pairs.

Neural Geodesic > Shortest Path



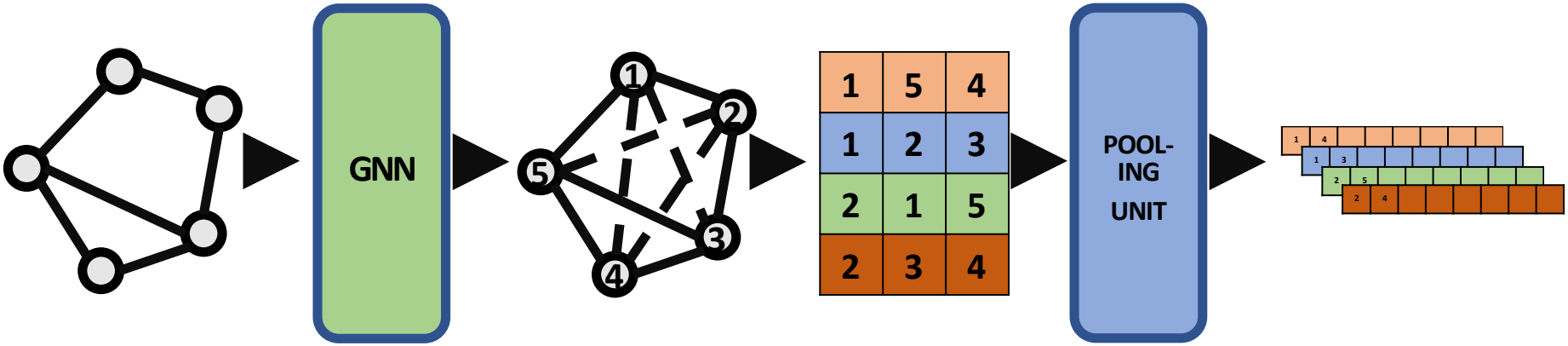
While A and C are symmetric, the brown node and green node are vastly different



$$e_{\blacksquare} \neq e_{\blacksquare} \rightarrow h_{A,B} \neq h_{C,B}$$

Using the Neural Geodesic is able to extend the expressiveness of shortest path

Geodesic GNN



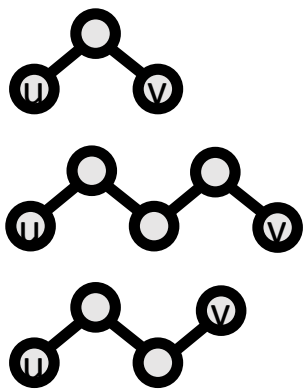
GNN applied only once to a homogeneously labeled graph

GNN generates embeddings for each node

Node embeddings on shortest paths are extracted

Pooling unit is applied to the embeddings to form the final representation of the link along with the head and tail embeddings

Summarize Geodesic by POOLING layer



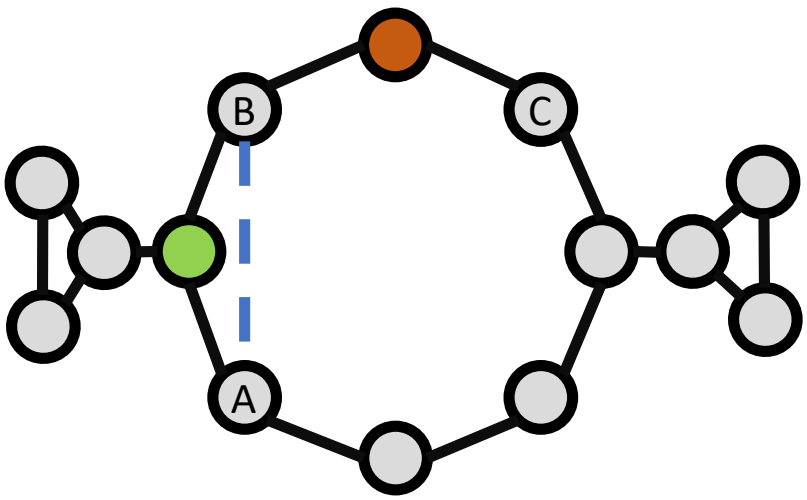
Geodesics can have variable lengths

Pooling Methods	Formula
SUM POOLING	$s_{P_{uv}} = \sum_{e \in E_{uv}} e$
Deepset POOLING	$s_{P_{uv}} = \sum_{e \in E_{uv}} MLP(e)$
RNN POOLING	$s_{P_{uv}} = RNN(E_{uv})$

P_{uv} is the geodesic. E_{uv} is the set of node embedding on the geodesic

$s_{P_{uv}}$ is the pooled neural geodesic representation.

Final link representation

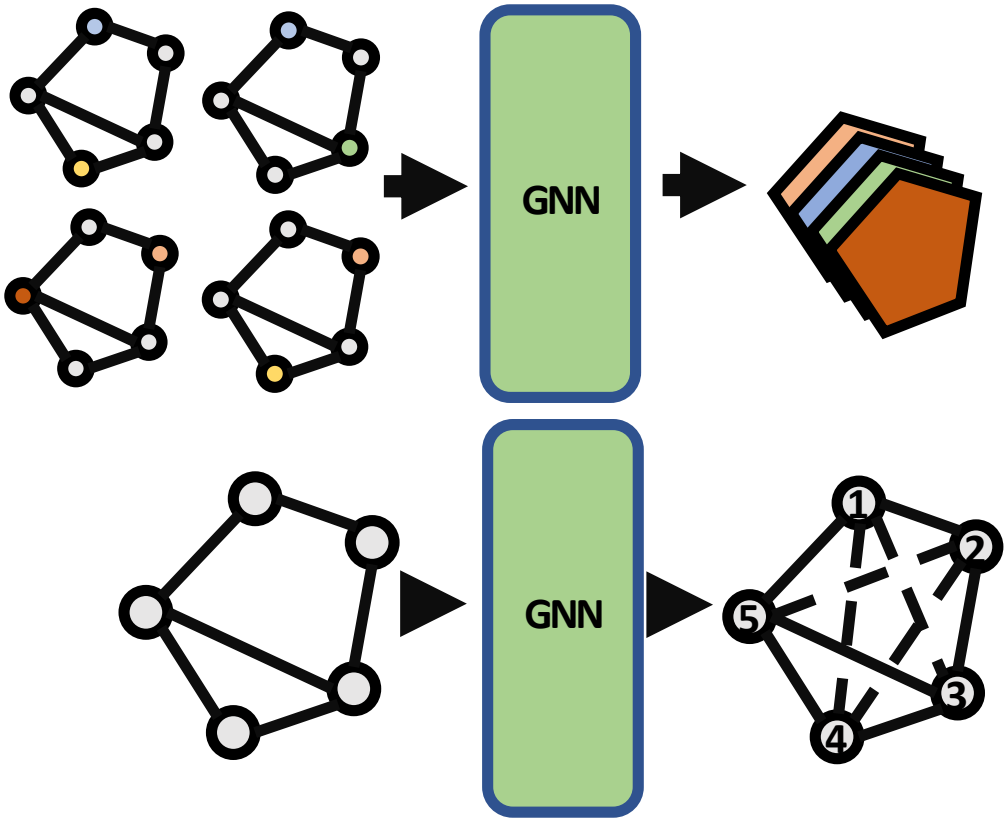


For the blue dotted link connecting AB, the link representation is:

$$h_{AB} = [e_A \oplus e_B \oplus s_{P_{AB}} \oplus dist(A, B)]$$

Distance is also used to speed up and regularize training.

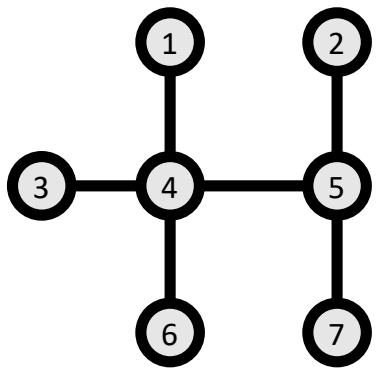
Geodesic GNN is very efficient



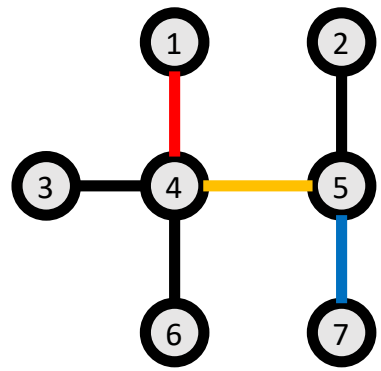
Labeling methods require 1 GNN run for every link of interest

Geodesic GNN only runs the GNN once, and the POOLING layers are very fast

Knowledge Graphs



$$a_4 = AGG(E_{[1,3,5,6]})$$



$$a_4 = AGG(Re_1, Re_3, Re_6, Re_5)$$

Inductive knowledge graph link prediction

Model trained/validated on a knowledge graph.

The model is tested on a graph with completely unseen entities.

Requires the model to learn instead of remembering the graph structures.

	FB15K-237		WN18RR	
	v1	v3	v1	v3
DRUM	0.529	0.529	0.744	0.462
NeuralLP	0.529	0.529	0.744	0.462
GraIL	0.642	0.828	0.825	0.584
NBFNet	**0.834**	0.951	0.948	0.893
GD-GNN-DS	0.800	**0.973**	0.931	0.889
GD-GNN-LSTM	0.771	0.957	**0.952**	**0.896**
GD-GNN-SUM	**0.829**	0.966	0.947	0.836

Draw 50 negative links, rank the probability of the test link against the negative links, report the Hit@10 rate.

Transductive graph link prediction

Model trained/tested on a homogeneous graph.

Because the train and test graph are the same, model with ability to remember the node identity will have a better performance.

	Cora		PubMed	
	AUC	AP	AUC	AP
VGAE	0.914	0.926	0.944	0.947
S-VGAE	0.941	0.941	0.960	0.960
SEAL	0.933	0.942	0.978	0.979
NBFNet	**0.956**	**0.962**	**0.983**	**0.982**
GD-GNN-DS	0.918	0.926	0.964	0.965
GD-GNN-LSTM	0.899	0.908	0.976	0.977
GD-GNN-SUM	0.924	0.936	0.947	0.944

Draw 1 negative links for each test sample using the probability of the links as scores to calculate AUC and average precision score.

Time complexity

Because GD-GNN requires only 1 GNN run, the more links to be evaluated the less the amortized running time, and hence GD-GNN is more efficient.

	Transductive					Inductive	
	FB15K-237	WN18RR	Cora	CiteSeer	PubMed	FB15K-237	FB15K-237
# of links	20466	3134	544	474	4435	2972	2663
GraIL	$\approx$ 1 month	$\approx$ 10 days	-	-	-	7 mins	9 mins
NBFNet	4 mins	2.4 mins	$<$ 1 sec	$<$ 1 sec	25 secs	6 secs	20 secs
GD-GNN-SUM	1.7 mins	1 min	$<$ 1 sec	$<$ 1 sec	5 secs	6 secs	7 secs

Comparison with GNN-based methods that achieved comparable performance

Ablation study

GD-GNN is a generic framework that can be applied to any existing GNN.

Neural geodesic significantly improves the performance of using distance alone.

	GCN		GIN		GraphSAGE	
	AUC	AP	AUC	AP	AUC	AP
No Geodesic	0.694	0.724	0.624	0.661	0.575	0.541
W/ Distance	0.843	0.882	0.614	0.642	0.832	0.865
W/ Geodesic vector	0.903	0.913	0.865	0.870	0.888	0.898

Experiments conducted on Cora dataset

Limitation and future direction of GD-GNN

- 1, GD-GNN is not an universal estimator.
- 2, GD-GNN is not performing well in transductive knowledge graph task.

	FB15K-237				WN18RR			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
DRUM	0.343	0.255	0.378	0.516	0.486	0.425	0.513	0.586
NeuralLP	0.240	-	-	0.362	0.435	0.371	0.434	0.566
TransE	0.294	-	-	0.465	0.226	-	-	0.501
RotatE	0.338	0.241	0.375	0.553	0.476	0.428	0.492	0.571
HAKE	0.346	0.250	0.381	0.542	0.497	0.452	0.516	0.582
RGCN	0.273	0.182	0.303	0.456	0.402	0.345	0.437	0.494
NBFNet	**0.415**	**0.321**	**0.454**	**0.599**	**0.551**	**0.497**	**0.573**	**0.666**
GD-GNN-DS	0.212	0.131	0.230	0.370	0.462	0.393	0.486	0.591
GD-GNN-LSTM	0.251	0.170	0.266	0.434	0.431	0.343	0.481	0.585
GD-GNN-SUM	0.204	0.118	0.217	0.353	0.432	0.352	0.472	0.580

Thank you!